Roads and eucalyptus forest expansion, 2018–2025

Aim and predictions

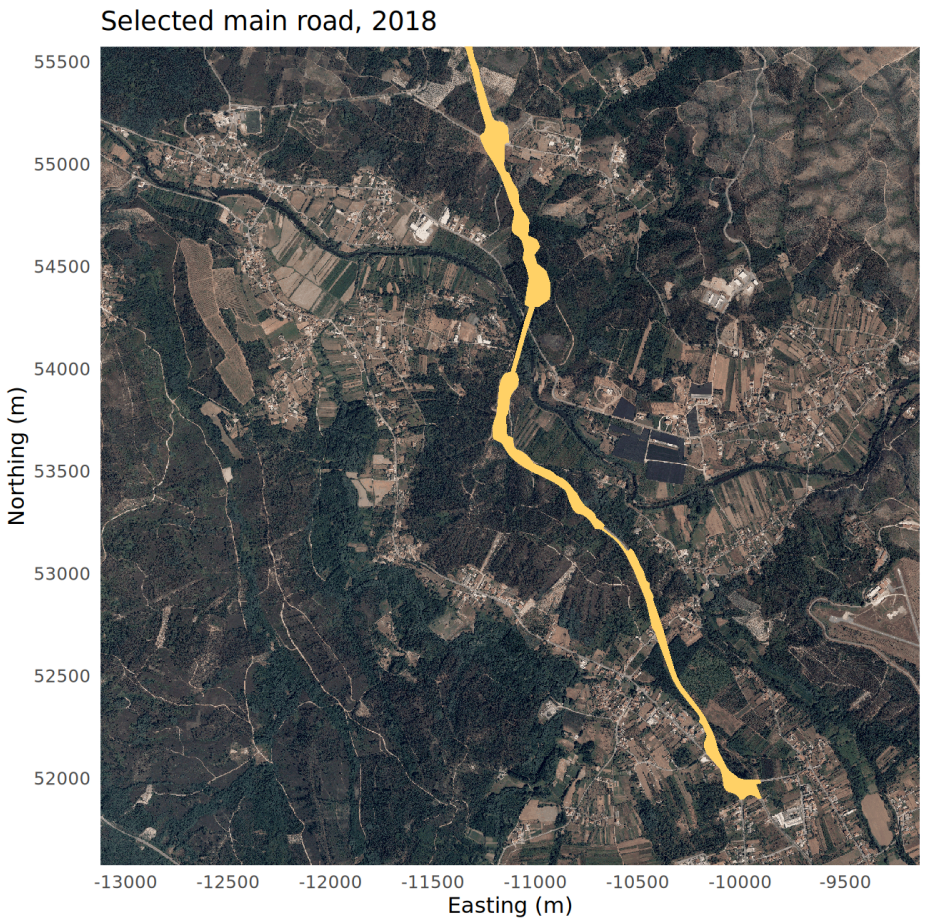
We ask whether the road network is associated with eucalyptus forest expansion in central Portugal. We predict (1) more new eucalyptus cover near roads and (2) a slower decline in new cover with distance from old eucalyptus near roads. The current results combine all included road types into one distance measure. The first maps document the pilot and regional preparation; the poster figures and model results follow.

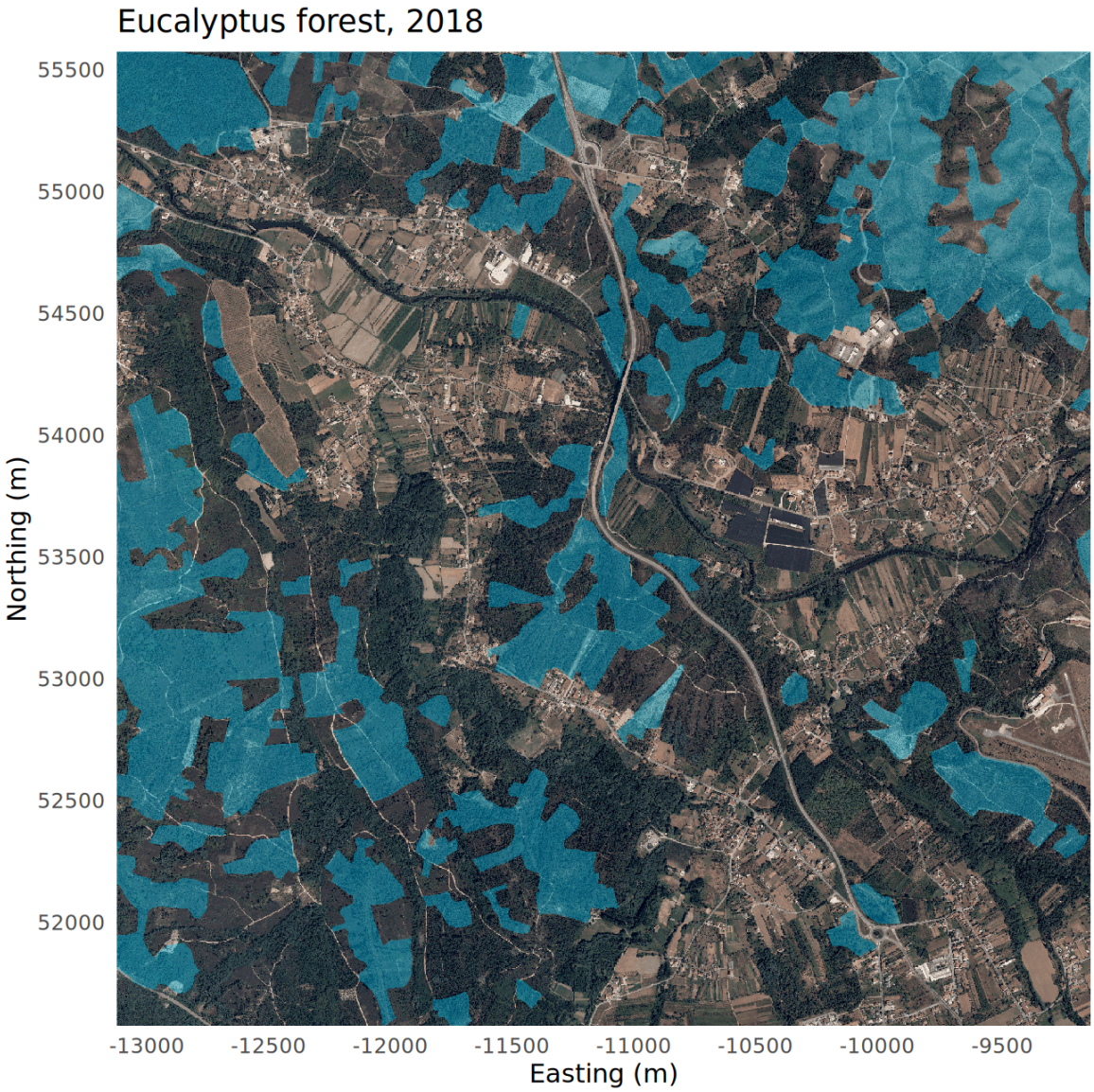
Pilot study in area B

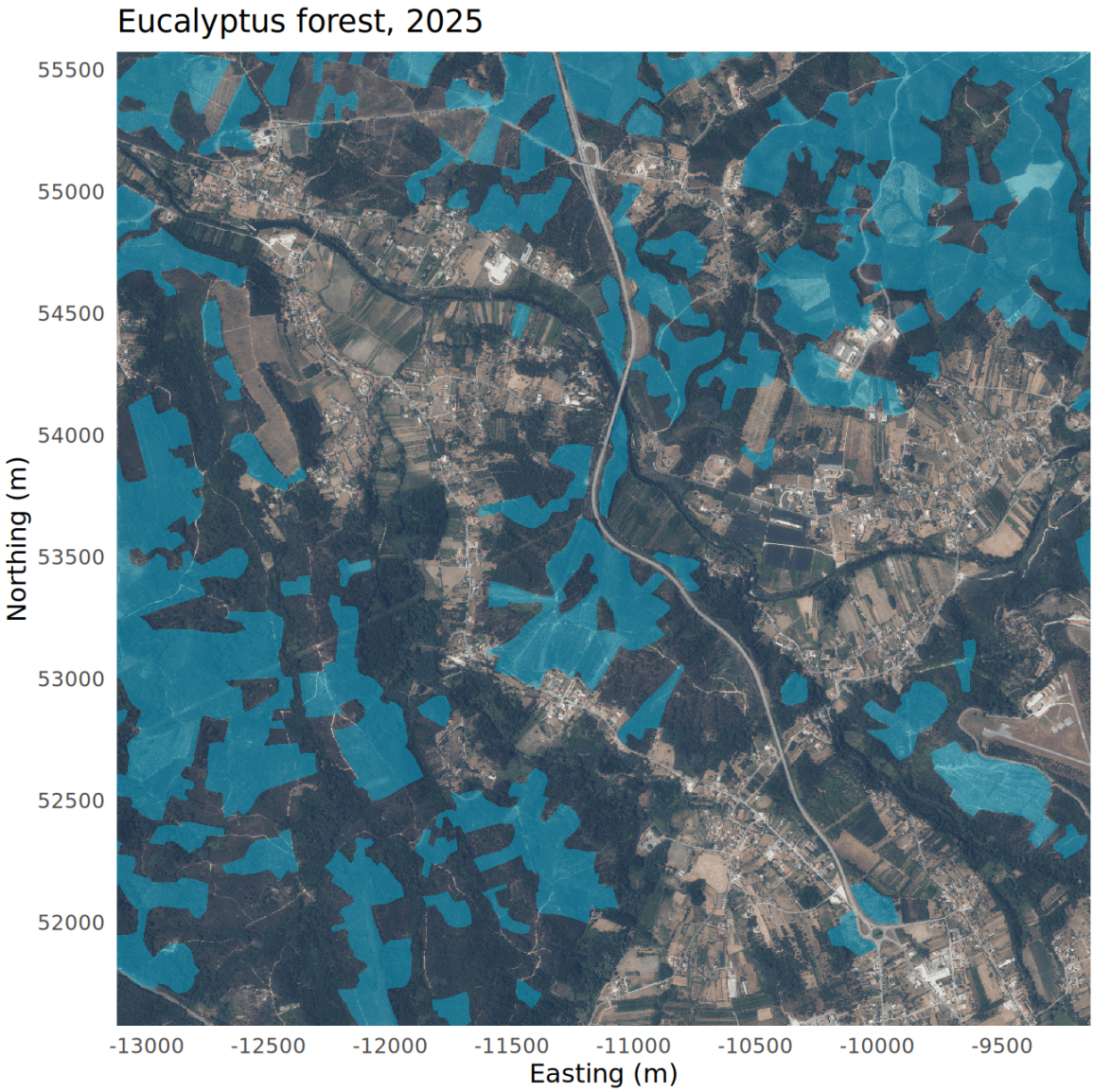
We first looked at a 4 km by 4 km area near Coimbra, Portugal. The yellow line on the first map shows the main road we selected.

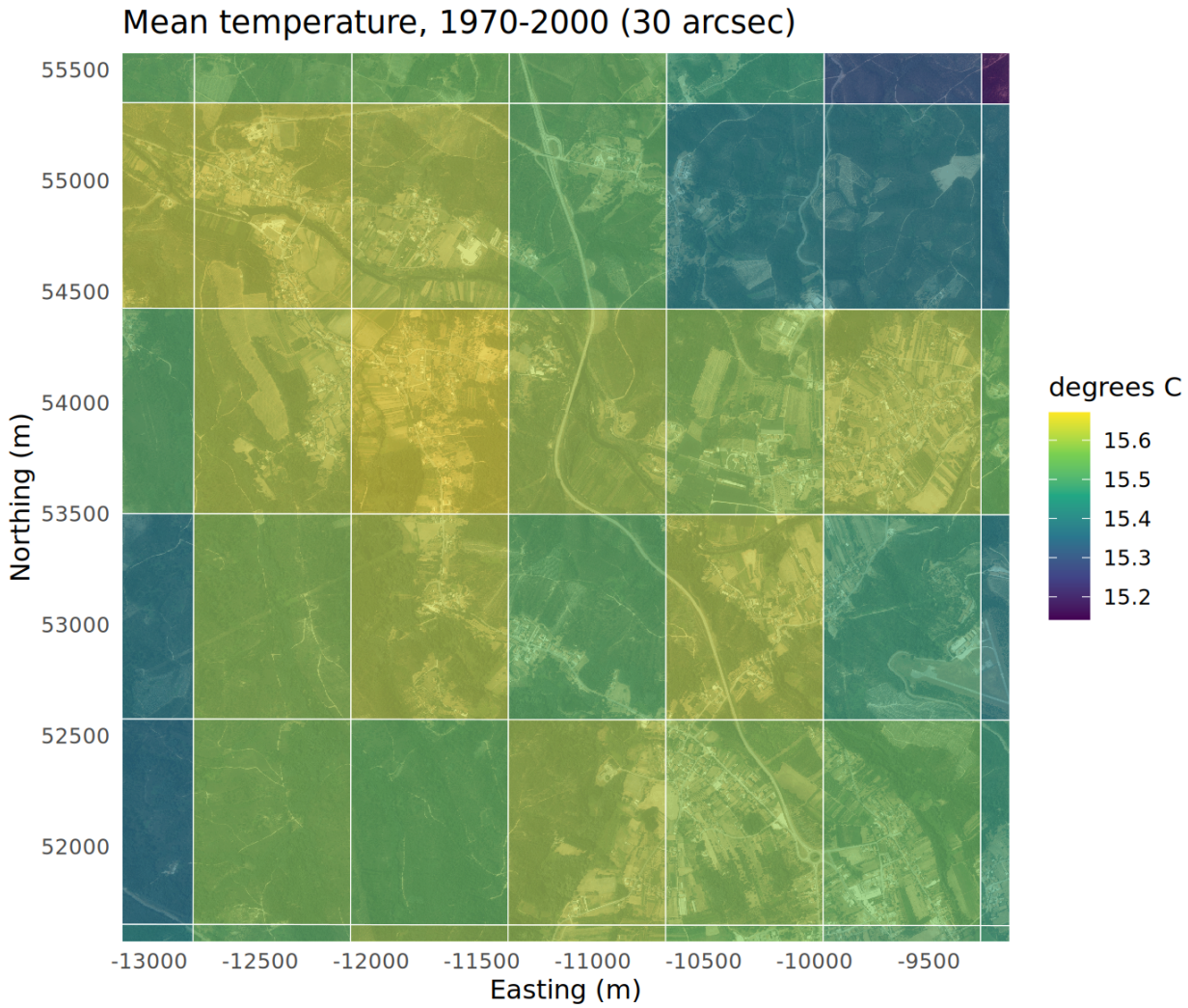
Four fields from the 2018 land-cover map are shown below.

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Rows: 6,063
Columns: 4
$ id         <int> 1288, 1294, 1296, 1297, 1299, 1323, 1361, 1367,~
$ cover_code <chr> "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.1.1.1", "1.~
$ cover_name <chr> "Áreas edificadas residenciais contínuas predom~
$ area_ha    <dbl> 2.5112068, 4.9795026, 1.3266062, 7.0387354, 2.8~
```

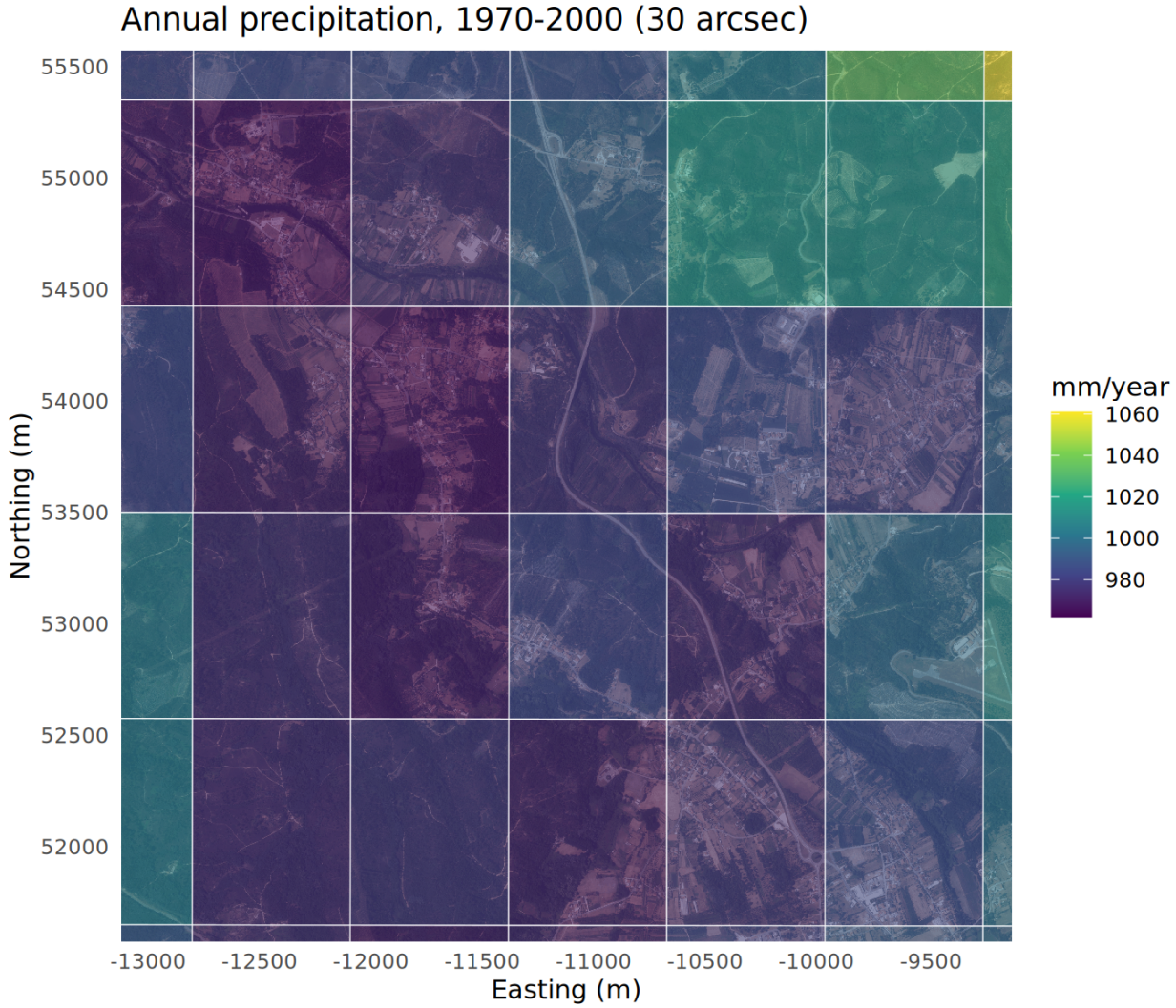

Selected main road, 2018
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)


Eucalyptus forest, 2018
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)

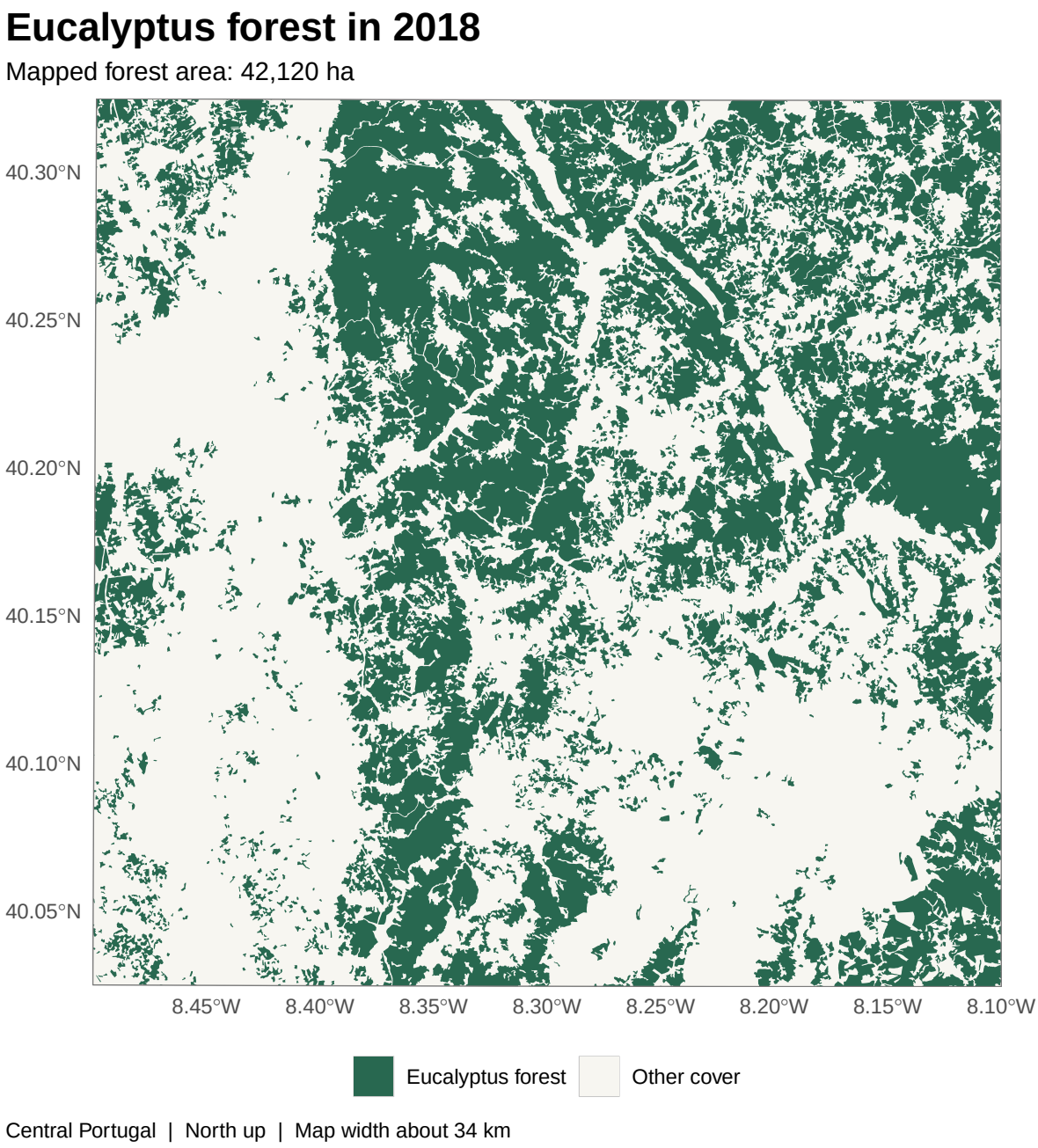

Eucalyptus forest, 2025
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)



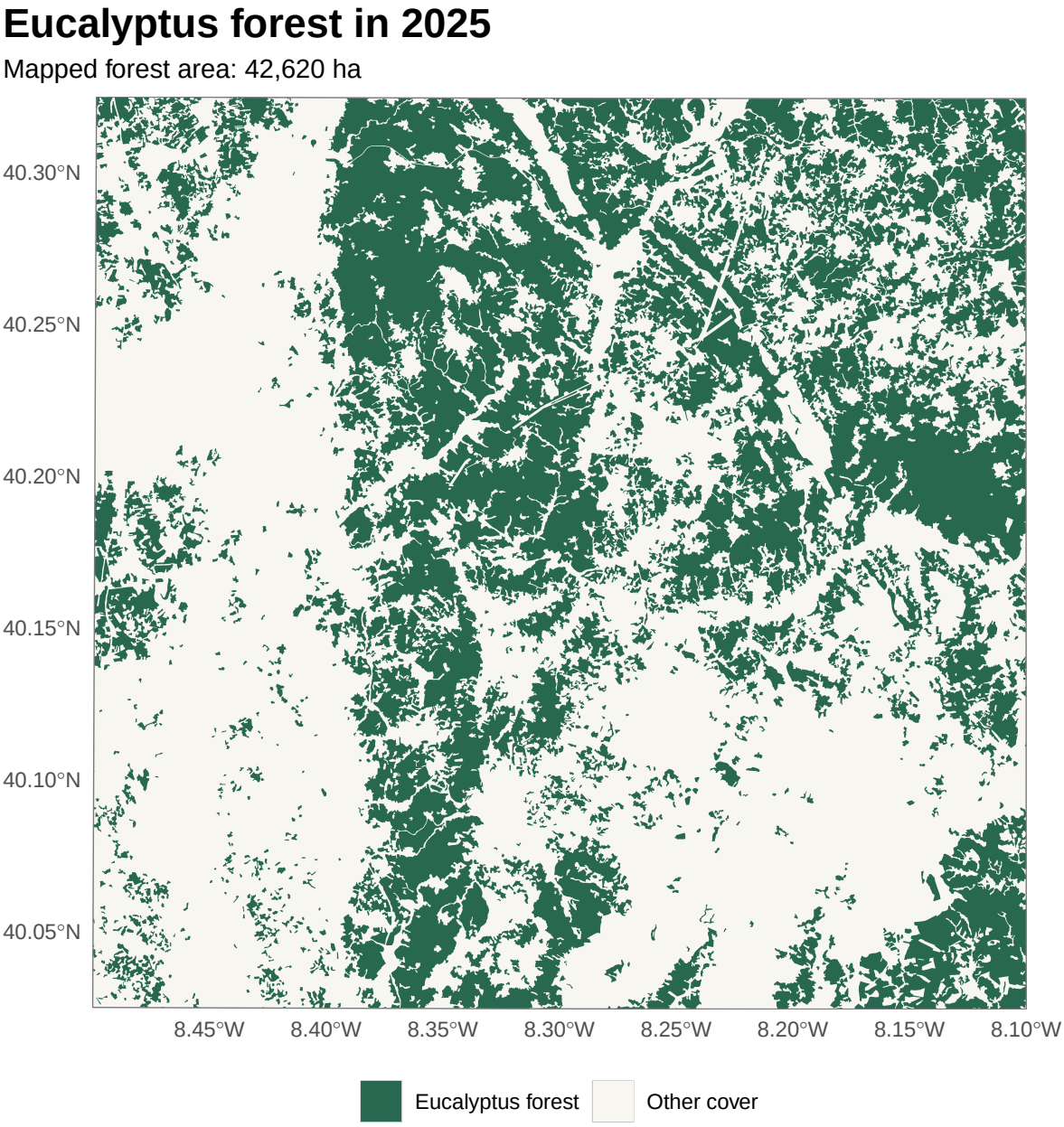
The climate maps show average temperature and annual rainfall for 1970–2000 over the 2025 aerial photo. Each climate cell covers about 710 by 925 m here.


Annual precipitation, 1970-2000 (30 arcsec)
55500
55000
54500
54000
53500
53000
52500
52000
Northing (m)
-13000
-12500
-12000
-11500
-11000
-10500
-10000
-9500
Easting (m)
mm/year
1060
1040
1020
1000
980

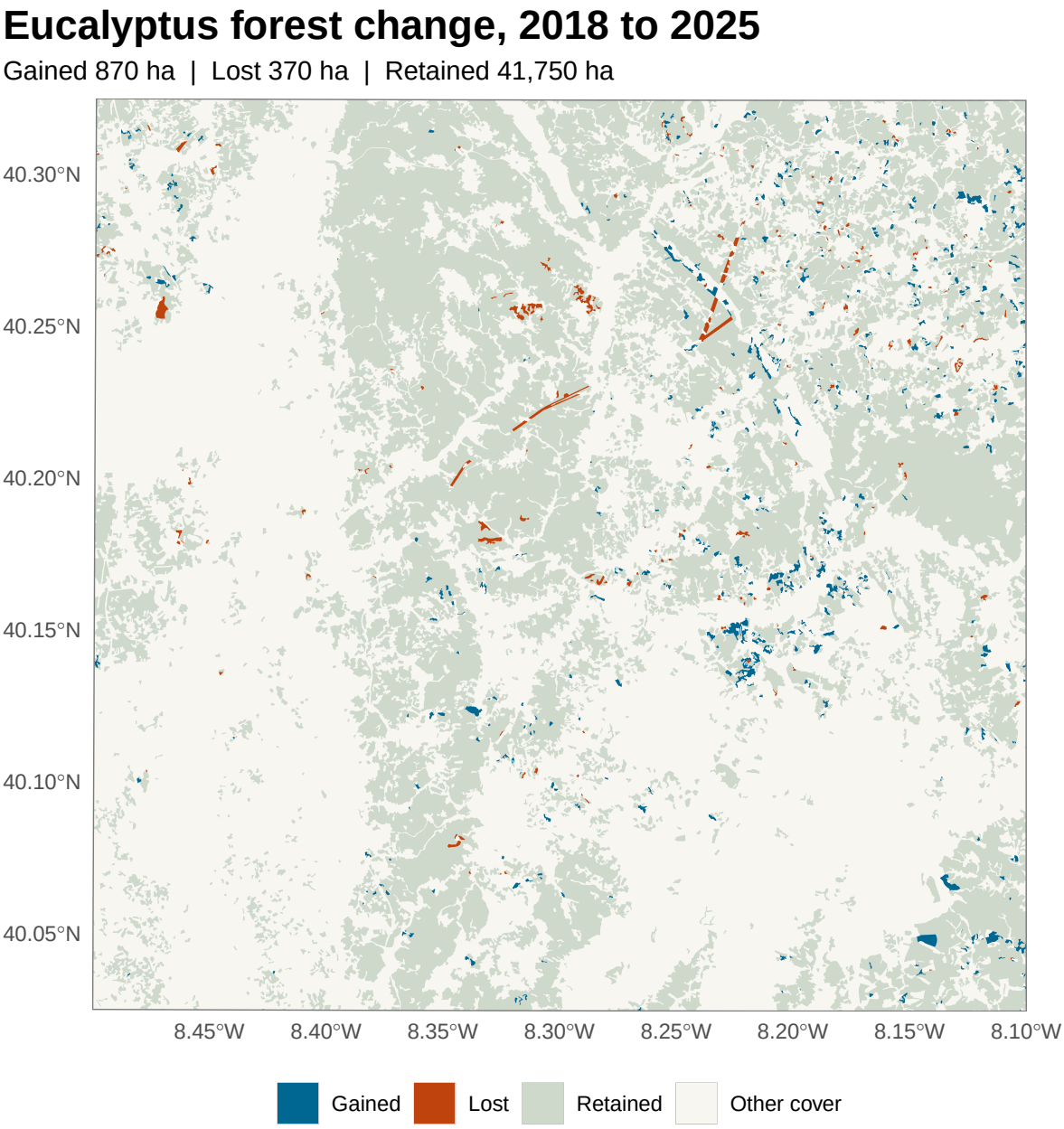
Regional eucalyptus forest cover



Green shows forest mapped as mainly eucalyptus in 2018. The study area is near Coimbra in central Portugal (8.50–8.10°W, 40.025–40.325°N).



The 2025 map shows eucalyptus forest in the same area and at the same scale.



Blue shows eucalyptus forest gained between 2018 and 2025. Orange shows forest lost, and pale green shows forest mapped in both years. The net gain was **500.3 ha**. These maps show the main land cover in each area.

Environment table

Each row in `environment.csv` is one point on the 80 m grid. This was our first environmental overview. The updated comparisons below use actual land areas, organised in 20 m cells. Nearby locations can share the same climate value.

Updated roads

We kept the comparable **2018 and 2025** forest maps and rebuilt the roads from the **January 2018** OpenStreetMap archive. We now include 17,113 mapped line segments in the study area and its distance buffer. Segments are not separate whole roads.

Group	Included routes	Segments
Traffic roads	Motorways, trunk, primary, secondary and tertiary roads and their links; minor connecting, residential and living-street roads	11,308
Land-access routes	4,206 tracks plus 1,599 service segments, including driveways, alleys and service routes without a subtype	5,805

Private and agricultural access restrictions do not remove a physical route, so these routes remain. We exclude parking aisles, drive-through lanes, footpaths, cycleways, steps, pedestrian streets, raceways, unknown road types and unfinished routes. This is a definition for this study; it does not mean the excluded routes have no ecological effect.

A service road is not necessarily a forest road. Of the 1,182 service segments without a subtype, 48% of their length inside the study area crosses artificial surfaces in the 2018 map, and about 26% crosses forests. We therefore call the second group land-access routes. We check results with only tracks, without unknown service subtypes, and without the 29 segments explicitly tagged as closed to motor vehicles.

For the poster analysis, distance means the nearest included route across both groups: the smaller of traffic-road and land-access distance. All distances enter one continuous model. The archive can miss routes and does not measure traffic or the exact construction date.

Our original predictions. Result 1 asks whether comparable land gains more eucalyptus near roads. Result 2 asks whether gain declines less with distance from old eucalyptus near roads. Both test the broader hypothesis that roads favour forest expansion. The burned-site maps provide an additional exploration of possible explanations.

How we compare land

Each initially non-eucalyptus fragment is assigned to its nearest 2018 eucalyptus source feature and its starting land-cover class. We compare new cover as a fraction of eligible land area, using area weights. Large amounts of far-road land therefore do not receive the same denominator as a small amount of near-road land. Source-by-cover groups have their own intercepts; terrain, climate, wind, fire history and management are included in the model. This is statistical adjustment, not one-to-one matching.

The 20 m cells are accounting units, not individual trees or independent samples. The main window is 100 m from old forest. It defines the nearby forest-edge landscape; it is not a measured seed-flight distance. The 25–75 m contrast describes change inside this window. Road references are the baseline eligible-area quartiles, selected without using the gain outcome. Confidence intervals allow shared source features and shared 2 km spatial blocks; 2 km is an uncertainty grouping, not a dispersal distance.

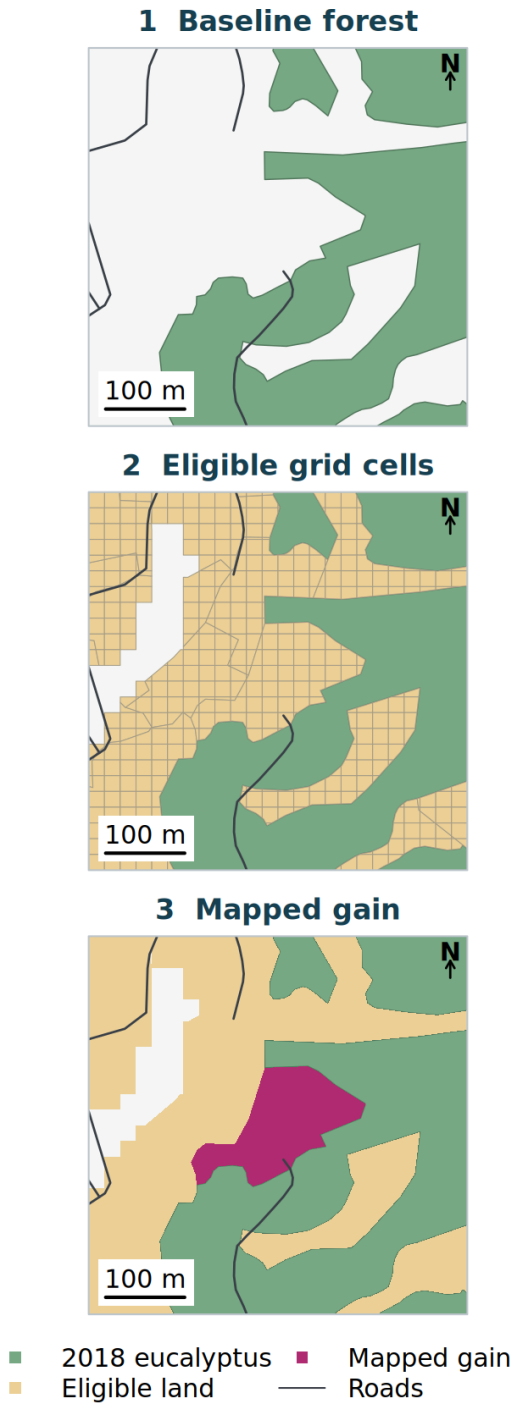


Figure 1: The same map window in three steps: baseline eucalyptus, eligible land divided into 20 m accounting cells, and mapped eucalyptus gain. The cells keep their actual eligible area; they are not individual trees. This example illustrates the workflow and is not an effect estimate.

The complete-environment model contains **26,309 ha** of eligible land and **766 ha** of mapped gain. The near and far road references are **36 m** and **169 m**, respectively. Middle distances remain in the continuous model. Result 1 is reported for source/cover groups spanning both road references. Result 2 also requires source distances to span 25–75 m at both road-distance ends.

Result 1: more gain near roads?

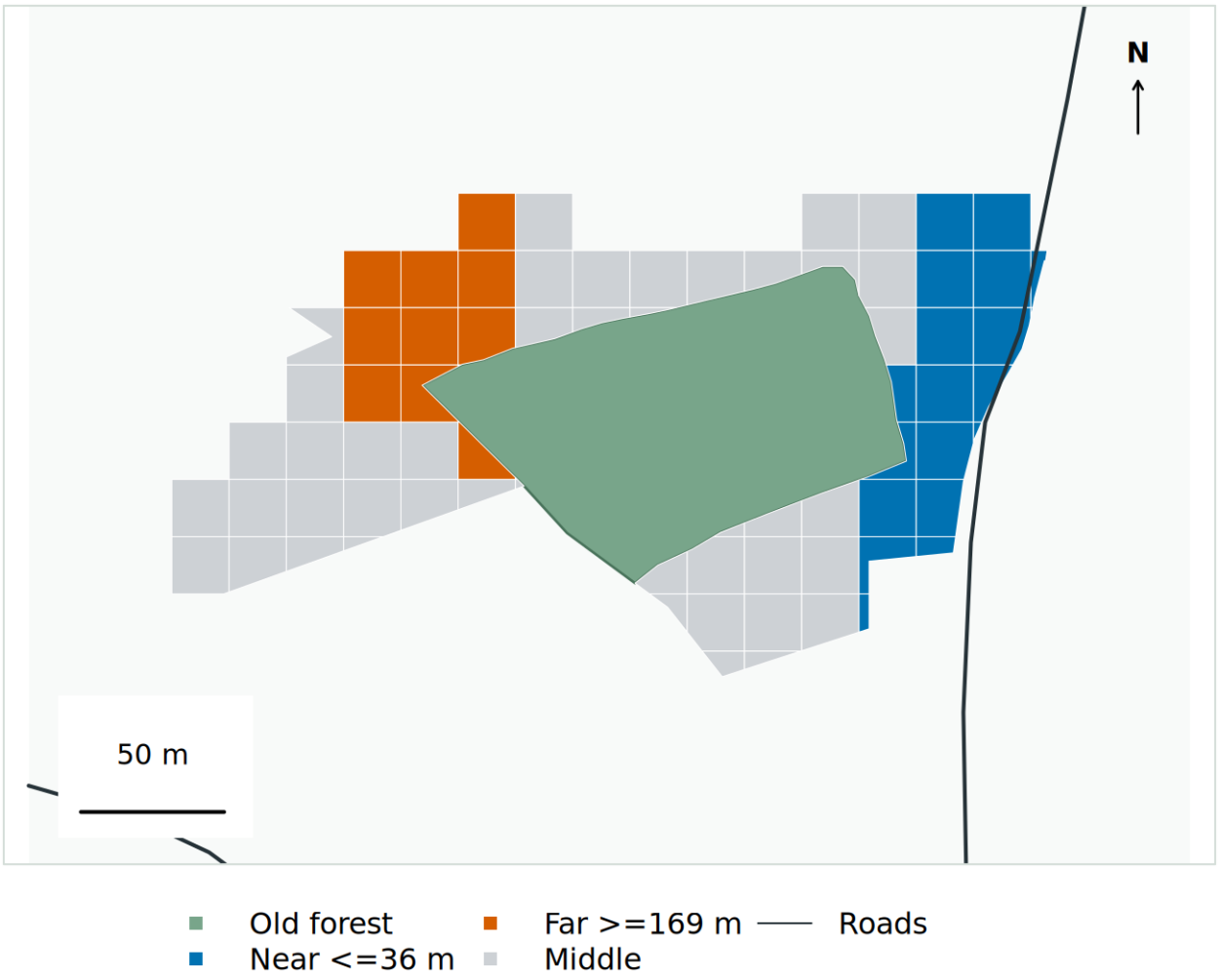


Figure 2: One baseline eucalyptus patch and eligible maritime-pine land within 100 m. Blue and orange show land closer to and farther from the included road network, using the baseline area-weighted 25th and 75th percentiles (about 36 and 169 m). Grey land also enters the continuous-distance model. This is an illustration of the comparison, not a separate local result or exact environmental matching.

Across the supported land, the adjusted near-minus-far difference was **+0.03 percentage points** (95% CI: -0.64 to +0.69). The comparison covers 778 source features and 17,760 ha. This does not clearly support more gain near roads. The map illustrates the comparison around one source; the estimate comes from the regional model.

Result 2: a slower decline near roads?

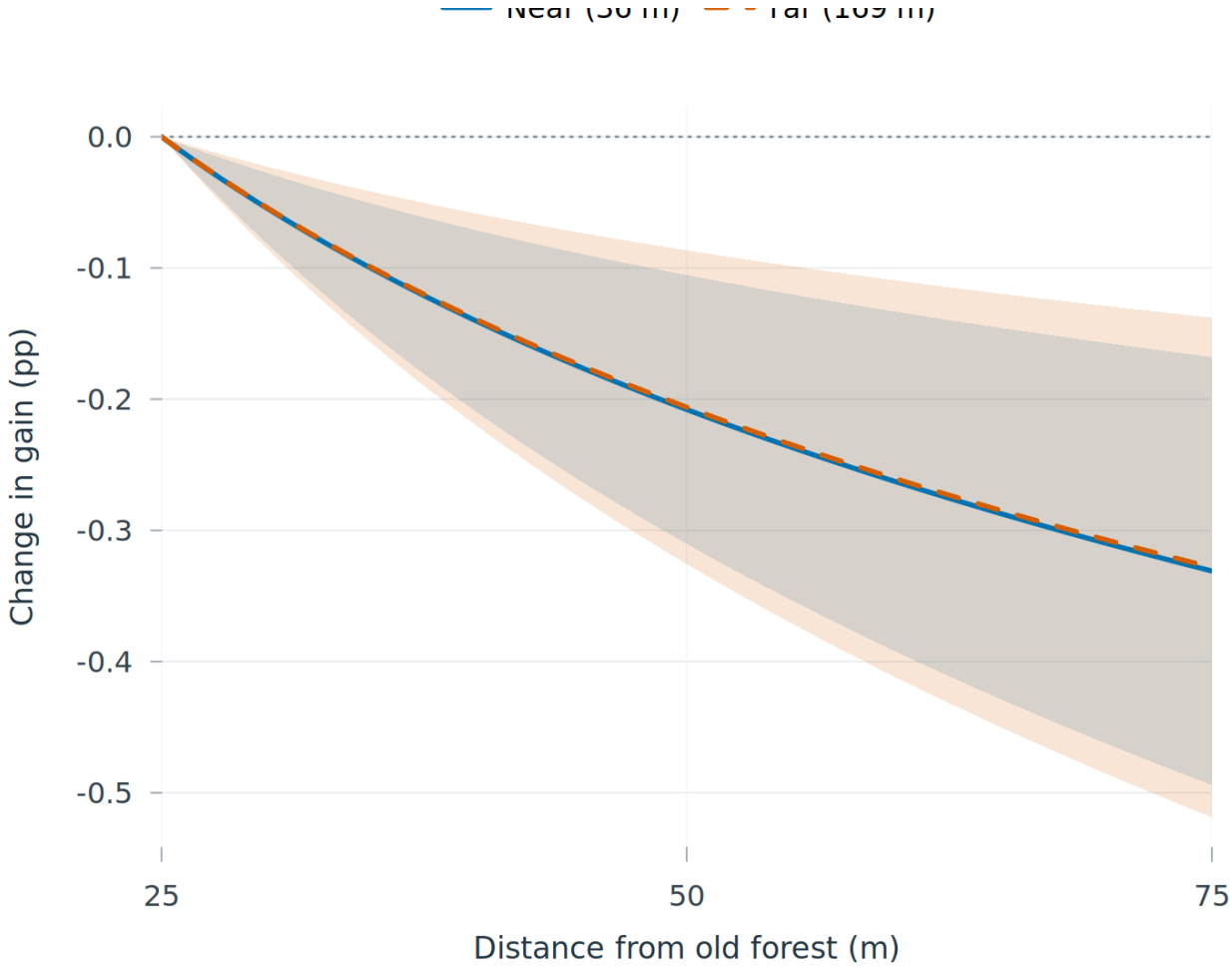


Figure 3: Adjusted change in mapped gain relative to 25 m from old forest, at near and far road references. Shading shows 95% confidence intervals. Both curves start at zero by definition; they are changes, not absolute probabilities.

Gain declined at nearly the same rate near and far from roads as source distance increased from 25 to 75 m. The near-minus-far difference in that change was **-0.003 percentage points** (95% CI: -0.130 to +0.124). The supported population contains 243 source features and 12,497 ha. This does not support a slower decline near roads. The interval is calculated for the contrast directly, rather than judged from overlap between the shaded bands.

Discussion: burned-site examples

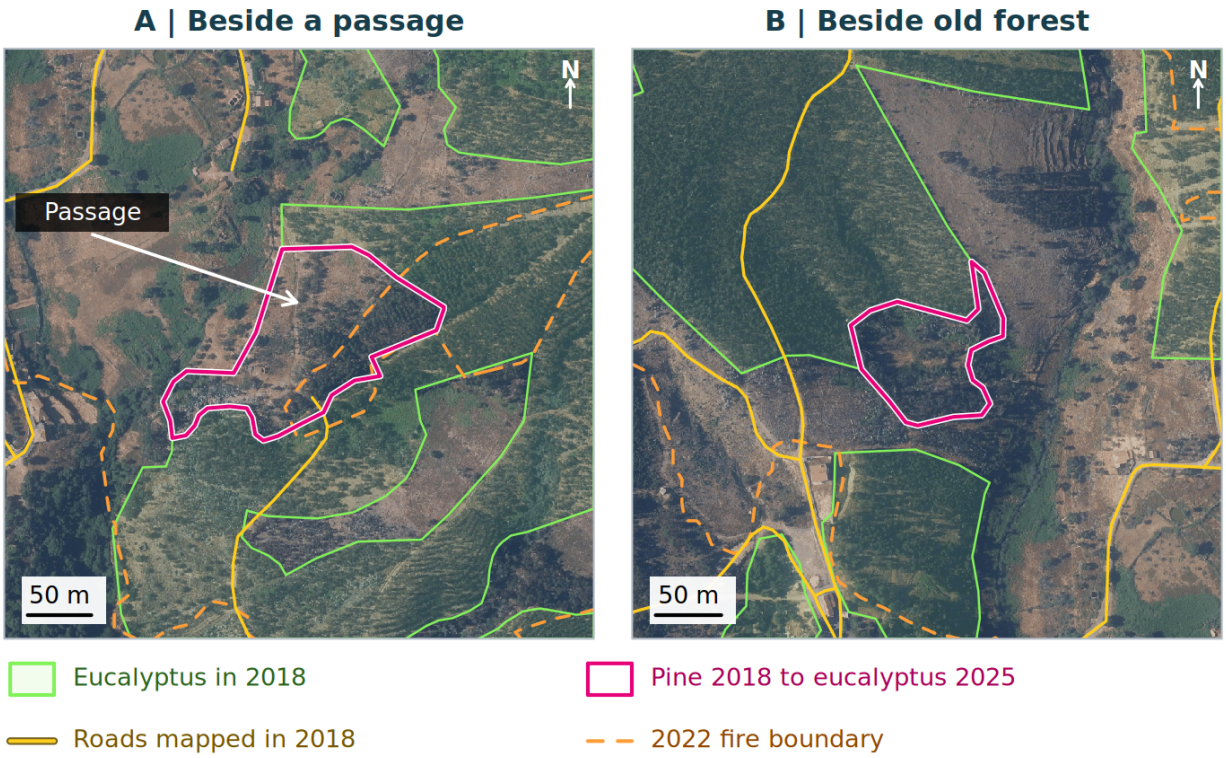


Figure 4: Two exploratory sites on unchanged 2025 aerial images. Pink outlines show whole polygons mapped as maritime pine in 2018 and eucalyptus in 2025; green outlines show baseline eucalyptus. Site A is beside a visible passage and only partly overlaps the 2022 fire. Site B is beside old eucalyptus. These images do not establish natural spread, planting history, or whether conversion occurred after the fire.

Both sites were mapped as maritime pine in 2018 and eucalyptus in 2025. Site A lies beside a visible passage and partly overlaps the 2022 fire footprint; site B lies beside old eucalyptus. Both were also within the 2017 fire footprint. These are selected examples for examining alternative explanations, not an independent test of a road effect.

Nearby seed sources, natural establishment, surviving trees and regrowth, planting, and post-fire management could each produce the observed forest changes. The maps do not date the conversion within 2018–2025, so the 2022 fire cannot automatically be treated as its cause. Neither location has a verified planting history. Coarse management labels cannot resolve this at individual patches. Agricultural abandonment, soils, drainage, road maintenance and actual traffic are further possible influences that are not fully measured.

Neither original road prediction was supported by this analysis. This does not establish that roads have no effect: the observations concern mapped forest-cover change, not tracked seed movement or confirmed natural colonisation. Dated establishment and planting records would help separate natural spread from forest management and recovery.

Model checks and limits

The linear model puts **40.7%** of fitted land area outside the 0–1 range. Its contrasts should therefore not be read as reliable absolute probabilities. We compare the same linear model with a bounded

fractional model on groups containing both changed and unchanged land. All zero- and complete-gain fragments inside these retained groups stay in the analysis.

Model	Result 2 (pp)	95% interval (pp)	Supported area (ha)	Source features
Linear: all eligible land	-0.003	-0.130 to 0.124	12497	243
Linear: informative groups	-0.209	-0.602 to 0.184	6062	60
Bounded: same informative groups	-0.142	Not calculated	6062	60

Changing the included source/cover groups also changes the linear estimate. The bounded fit uses that same smaller population and gives a point estimate without a confidence interval. It does not establish either robust support for a road effect or equivalence between near and far land.

The official maps describe dominant cover classes, including young stands, rather than individual trees. Classification changes and mixed stands can contribute to a mapped transition. A 20 m accounting grid does not improve the source map's detail, and climate cells remain much larger than these accounting cells.

Reproduction and software

Run `quarto render analysis.qmd` from the repository root after installing the packages listed in README. The spatial and model inputs use relative `Raw Data/` paths. The WorldClim pilot file downloads automatically on its first use. Exact package versions are listed in README; this chunk records the current rendering session in the ignored cache. No random sampling is used, and the seed is fixed at 6130.

Data sources

- **Land cover and the pilot road.** Direção-Geral do Território (DGT), COS Series 2, 2018 v4 and 2025 v1. The maps use eucalyptus class 5.1.1.6 and maritime pine class 5.1.2.1. The regional `pine2025.gpkg` file contains the 2025 maritime-pine polygons clipped to the same study area. DGT class definitions and imagery dates. DGT, CC BY 4.0
- **Regional roads.** The original network is retained in `roads.gpkg`. The revised `roads2018_selected.gpkg` comes from the 1 January 2018 archive and records traffic roads, tracks and selected service roads; the current model combines these into one distance. Private access is retained. © OpenStreetMap contributors. Geofabrik archive; track, service and access definitions; ODbL 1.0.
- **Pilot aerial photos.** DGT imagery from 2018 and 2025, exported at 2 m for area B. DGT orthophotos, CC BY 4.0
- **Pilot climate.** WorldClim v2.1, temperature and rainfall averages for 1970–2000. WorldClim · Fick and Hijmans (2017)
- **Regional climate.** CHELSA v2.1, temperature, rainfall and mean near-surface wind speed for 1981–2010. CHELSA · BIOCLIM+ data, CC0
- **Elevation and slope.** Copernicus DEM GLO-30, tile N40/W009. Product and licence
- **Fire.** ICNF, *Áreas ardidas*, 2017–2025, retrieved 5 November 2026. Fire perimeters · ICNF terms. The 2017 burn severity and image quality layers come from Jahanianfard et al. Portuguese Burn Severity Atlas v2, CC BY 4.0
- **Forest management.** Lesiv et al., global forest management map for 2015, v3.2. Data, CC BY 4.0 · Map validation and limits
- **Planting records.** ICNF public RJAAR projects and national summary of approved actions.

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The poster maps use `poster_map_inputs.gpkg`, a small derived snapshot of the reviewed map inputs, and two unchanged 2025 orthophoto PNGs. The GeoPackage keeps the real 20 m eligible fragments, baseline forest, complete case-transition polygons, selected roads, map frames, and the actual fire perimeter. It contains no new field observations. The map code reads these files directly from `Raw Data`; no files from the working folder are needed.

Forest classes are from DGT COS2018v4 and COS2025v1. The aerial images are from DGT orthophotos. DGT data are credited under CC BY 4.0. Roads are from OpenStreetMap contributors, under ODbL 1.0. Fire perimeters are from ICNF, subject to ICNF reuse terms. These source licences still apply to the derived map inputs; the repository's code licence does not replace them.

Research papers

- **Road corridors.** Lemke et al. (2019). Journal of Applied Ecology.
- **Vehicle seed movement.** von der Lippe et al. (2013). PLOS ONE.
- **Field distance.** Deus et al. (2019). Wild eucalyptus at and beyond plantation edges in Portugal. Biological Invasions
- **Source distance in a mapping study.** Deus et al. (2022). Potential eucalyptus recruitment in Iberian protected areas. Forests
- **Fire and pine.** Águas et al. (2014). Natural establishment of eucalyptus in burned stands in Portugal. Forest Ecology and Management
- **Fire and resprouting.** Catry et al. (2013). Survival and regrowth of planted blue gums after fire in Portugal. Forest Ecology and Management
- **Pine recovery.** Maia et al. (2012). Fire severity and maritime-pine regeneration in central Portugal. Annals of Forest Science
- **Seedling habitat.** Fernandes et al. (2018). Habitat disturbance and wild trees around Portuguese plantations. Biological Invasions
- **Farm and forest history.** Nunes et al. (2023). Land-use change in Lousã, central Portugal. Quaestiones Geographicae
- **Statistical methods.** Papke and Wooldridge (1996); fixest 0.14.2 documentation.